

Homework-2

Packages

Question 1

```
install.packages("aRtsy")
```

install.packages installs the package to your device, whereas the library function tells R to use the specific package in the code below the library function

Question 2

```
library(aRtsy)
```

The Collatz conjecture is a famous unsolved math problem that asks whether two simple arithmetic operations will eventually transform every positive integer into 1. If the number n is even, you divide by 2, written as $f(n) = n/2$. If the number n is odd, you multiply by 3 and add 1, written as $f(n) = 3n + 1$.

Question 3

part a

```
canvas_collatz(  
  colors = c("blue"),  
  n = 10  
)
```

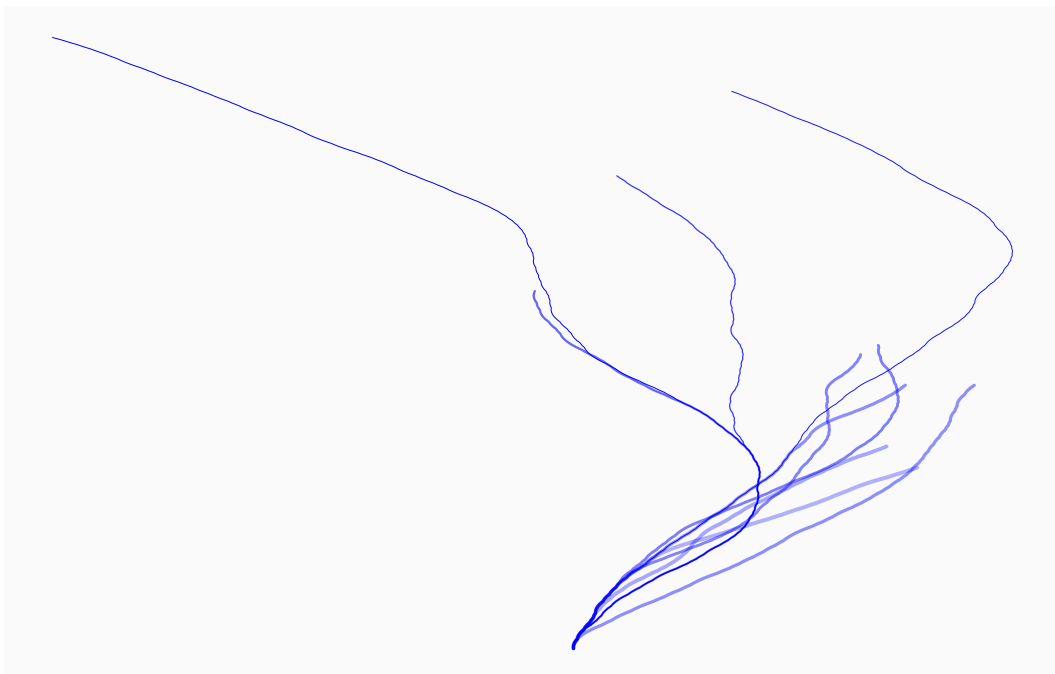
The above code uses the `canvas_collatz` function from the `artsy` package. It generates artwork based on the Collatz conjecture by calculating the starting paths for 10 randomly chosen starting integers. It plots these fractal paths on a canvas using the designated color.

part b

You get a different output every time because the code runs based off of 10 randomly generated integers. Since these 10 numbers change each time, every time you render this into a quarto file, it shows a different graph.

`Set.seed()` sets the seed for R's pseudo-random number generator. By setting a specific seed before running a function that picks random integers, R will generate the exact same sequence of pseudo-random numbers each time, ensuring that the output will be replicable across multiple renders.

```
set.seed(329386292)  
  
canvas_collatz(  
  colors = c("blue"),  
  n = 10)
```



Question 4

```
canvas_collatz(  
  colors = c("blue"),  
  n = 1:20  
)
```



This code is different than the previous code because it takes 20 specified integers instead of 10 random ones. This code chunk shows the graph of how the Collatz conjecture for the first 20 integers.

It would not be appropriate to use `set.seed(1)` in this code chunk because this code chunk evaluates the function for a set vector of 20 numbers without any random integers.